

Def. Let V be a vector space over a field F .

A function $f: V \times V \rightarrow F$ is called a bilinear form if: for all $\alpha_1, \alpha_2, \beta_1, \beta_2 \in V$, $c, d \in F$,

$$f(c \cdot \alpha_1 + d \cdot \alpha_2, \beta_1 + \beta_2) = c \cdot f(\alpha_1, \beta_1) + c \cdot f(\alpha_1, \beta_2) + d \cdot f(\alpha_2, \beta_1) + f(\alpha_2, \beta_2).$$

Def. Let V be a fin. dim vector space over F , and let $\mathcal{B} = \{\alpha_1, \dots, \alpha_n\}$ be a basis for V .

Given a bilinear form f , define the matrix of f in $\mathcal{B}$ as:

$$[f]_{\mathcal{B}} \stackrel{\text{def}}{=} (f(\alpha_i, \alpha_j))_{i,j}.$$

Thm. Let V be a finite-dimensional vector space over F , and $\mathcal{B}$ be a basis.

Then, the space $L(V, V, F) = \{f: V \times V \rightarrow F \mid f \text{ is bilinear form}\}$ is isomorphic to $M_{nn}(F)$.

Proof. Define $\mathcal{Q}: L(V, V, F) \rightarrow M_{nn}(F)$; $f \mapsto [f]_{\mathcal{B}}$. Denote $\mathcal{B} = \{\alpha_1, \dots, \alpha_n\}$.

Given $\alpha = x_1 \alpha_1 + \dots + x_n \alpha_n$ and $\beta = y_1 \alpha_1 + \dots + y_n \alpha_n$,

$$f(\alpha, \beta) = \sum_{i=1}^n x_i \cdot f(\alpha_i, \beta) = \sum_{i=1}^n x_i \cdot \sum_{j=1}^n y_j \cdot f(\alpha_i, \alpha_j) = \sum_{i=1}^n \sum_{j=1}^n x_i y_j \cdot f(\alpha_i, \alpha_j).$$

Thus, for $X = [\alpha]_{\mathcal{B}}$ and $Y = [\beta]_{\mathcal{B}}$, one $A = [f]_{\mathcal{B}}$,

$$f(\alpha, \beta) = X^t A Y = [\alpha]_{\mathcal{B}}^t [f]_{\mathcal{B}} [\beta]_{\mathcal{B}}.$$

Therefore, the bilinear form f is uniquely determined by the representation $[f]_{\mathcal{B}}$.

This proves the $\mathcal{Q}$ is bijection. And, the $\mathcal{Q}$ is linear, because

$$(c \cdot f + g)(\alpha_i, \alpha_j) = c \cdot f(\alpha_i, \alpha_j) + g(\alpha_i, \alpha_j), \text{ for any } c \in F \text{ and } f, g \in L(V, V, F).$$

Cor. If $\mathcal{B} = \{\alpha_1, \dots, \alpha_n\}$ is a basis for V , and $\mathcal{B}^* = \{f_1, \dots, f_n\}$ is the dual basis, then $f_{i,j}(\alpha_i, \beta) \stackrel{\text{def}}{=}} f_i(\alpha_i) \cdot f_j(\beta)$ ($1 \leq i, j \leq n$) form a basis for V^* .